



Faculty of Engineering
Machine Learning

**A MACHINE LEARNING AND LLM-BASED CHATBOT APPROACH TO GRD PREDICTION AND
KNOWLEDGE MODELING**

Authors:

Vicente Aravena Santa María.
Clemente Carreño Becerra.
Benjamín Huerta Mallea.
Joao Morales Zavaleta.
Renato Navarro Monje.

Professor:

Pablo Schwarzenberg

Santiago, Chile
2026

GRD Prediction from Clinical Data of Hospital El Pino: A Machine Learning Approach for Diagnosis-Related Group Classification

CINF104 - Machine Learning | Project 1, Phase 1 | Universidad Andres Bello

Academic Year 2026

Abstract

This paper presents a machine learning system for the automatic prediction of Diagnosis-Related Groups (DRG) using clinical discharge records from Hospital El Pino, Chile. The dataset contains 14,561 patient records with up to 35 ICD-10 diagnosis codes, 30 ICD-9-CM procedure codes, patient age and biological sex, spanning 526 distinct DRG categories. Three classification approaches were evaluated: Linear Support Vector Classifier (LinearSVC), Logistic Regression and a majority-class baseline, under a stratified validation framework. LinearSVC achieved the best performance, with a global accuracy of 62.9%, weighted F1 of 0.611 and Top-3 Accuracy of 76.9%. For the 20 most frequent DRG categories, weighted F1 reaches 0.85. The main challenge is the severe class imbalance across 526 categories, with a macro F1 of 0.29. Results are competitive with international literature on DRG prediction (range 55-75% accuracy) and demonstrate the viability of sparse linear models as a tool to support hospital coding.

Keywords: Diagnosis-Related Groups, DRG prediction, multiclass classification, LinearSVC, clinical coding, ICD-10, ICD-9-CM, hospital informatics.

I. INTRODUCTION

Diagnosis-Related Groups (DRGs) are a patient classification system that groups hospital discharges into clinically coherent categories with similar resource consumption patterns. Originally developed in the United States in the 1980s [1], the DRG framework has become the basis of prospective payment systems in numerous countries, including the Chilean IR-GRD system under FONASA.

Manual DRG assignment is a time-consuming and error-prone process performed by specialized medical coders. Incorrect coding can lead to underfunding for hospitals or incorrect billing, both with important financial and care quality implications. Automating this process through machine learning offers the potential to reduce coding errors, streamline hospital billing cycles and support real-time clinical decision-making.

This work addresses the challenge of predicting the DRG from the discharge information available in the Hospital El Pino dataset. The problem is formulated as a multiclass classification task with 526 possible categories. The input features are: ICD-10 diagnosis codes, ICD-9-CM procedure codes, patient age and biological sex — exactly the information available at the time of discharge.

The main contributions of this work are: (1) a systematic preprocessing pipeline for clinical data with multi-label code fields; (2) a comparative evaluation of machine learning models under a stratified validation framework; and (3) an analysis of model performance both globally and on the most frequent DRG categories, providing clinically interpretable conclusions.

II. LITERATURE REVIEW

Automatic DRG prediction from clinical data has been an active research area since the early 2000s. Early approaches applied rule-based systems that replicated the logic of the official DRG grouper, achieving high accuracy on structured data but with limited generalizability across hospital systems [2].

Systematic machine learning approaches began to appear with the greater availability of electronic health records. Goldstein et al. [3] demonstrated that decision tree classifiers trained on ICD codes could predict DRG membership with over 70% accuracy for a subset of high-volume categories. However, performance declined notably for infrequent categories — a central challenge in our work.

More recent studies have adopted deep learning techniques. Shi et al. [4] used a convolutional neural network on ICD-10 code embeddings to predict DRGs in a Chinese tertiary hospital, reporting weighted F1 of 0.72-0.81 depending on specialty. Baumel et al. [5] applied hierarchical attention networks to clinical notes for ICD coding, achieving state-of-the-art results, but requiring unstructured free text not available in many systems.

For structured tabular data (codes only), Support Vector Machines and logistic regression remain competitive baselines. Moons et al. [6] reported that LinearSVC with TF-IDF-weighted ICD code features achieved 68% accuracy across 312 DRG classes, consistent with our results. Their study also highlighted the importance of Top-k accuracy as a clinically relevant metric.

In the Latin American context, Soto et al. [7] analyzed the quality of DRG coding in Chilean hospitals and found systematic discrepancies between manual coding and grouper output, motivating the need for automated verification tools. To our knowledge, this is the first published machine learning study using Hospital El Pino data.

The key distinction of the present work is the extremely high number of target classes (526 DRGs) and the severe class imbalance, which places it among the most challenging DRG prediction tasks reported in the literature.

III. OBJECTIVE

The main objective of this study is to design, train and evaluate a machine learning model capable of predicting the DRG assigned to a patient at hospital discharge, using only the structured clinical information available in the administrative dataset of Hospital El Pino.

The specific objectives are:

- Conduct an exploratory data analysis to characterize dataset quality, including completeness, correctness and identification of outliers.
- Design a preprocessing pipeline that transforms multi-valued clinical codes into machine-learning-compatible feature vectors.
- Compare candidate classification models under a common stratified validation framework, selecting the best model based on macro F1.
- Analyze model performance both globally (526 classes) and on the most frequent classes (top 20), providing clinically actionable conclusions.

IV. METHODOLOGY

A. Dataset Description

The main dataset, `dataset_elpino.csv`, contains clinical discharge records from Hospital El Pino. It comprises 14,561 patient records with the following variable types:

- Clinical Variables: Up to 35 ICD-10-coded diagnosis columns and up to 30 ICD-9-CM-coded procedure columns. These codes have no predefined clinical ordering.
- Demographic Variables: Patient age (range 0-121 years) and biological sex (Female/Male).
- Target Variable: The assigned DRG code — a multiclass classification problem with 526 distinct categories.

Auxiliary reference files include: ICD-10.xlsx (39,873 ICD-10 descriptions), ICD-9.xlsx (4,646 ICD-9-CM codes), IR-GRD V3.1 with 2016 FONASA prices (1,095 DRG cost records) and master tables with DRG severity and mortality levels.

B. Machine Learning Techniques

Given the sparse and high-dimensional nature of multi-hot encoded clinical codes, we focus on linear models suited for this data structure:

- LinearSVC: Linear Support Vector Classifier with One-vs-Rest (OvR) strategy for the 526-class problem. Linear SVMs generalize well in high-dimensional spaces with sparse features [8].
- Logistic Regression (multinomial): Probabilistic baseline providing calibrated posterior probabilities, useful for computing Top-k accuracy. Trained with L2 regularization and a maximum of 1,200 iterations.
- Dummy Classifier (most frequent class): Always predicts the most frequent class — used as a lower baseline to confirm that learned models are non-trivial.

C. Evaluation Metrics

Model selection is based on the following metrics calculated on the validation set:

- Macro F1: The primary selection criterion. Computes the unweighted mean F1 across all 526 classes, penalizing models that ignore minority classes.
- Weighted F1: Weighted mean F1 (by class support), reflecting overall quality over the observed distribution.
- Global Accuracy: Proportion of correctly classified samples.
- Top-3 Accuracy: Fraction of samples where the correct DRG appears among the three highest-scoring predictions — clinically relevant since a coder reviewing 3 suggestions can validate the correct DRG much faster than a full manual search.

D. Model Development Process

The development pipeline follows these sequential steps:

- Data Cleaning: Removal of descriptive text columns; retention of code-only columns. Age clipping at the 1st-99th percentile range and median imputation for missing values.
- Feature Engineering: MultiLabelBinarizer applied separately to the set of non-null diagnosis codes per patient (up to 35 columns) and the set of procedure codes (up to 30 columns). OneHotEncoder applied to the sex variable.
- Feature Assembly: All feature matrices are horizontally stacked (sparse hstack) into a single high-dimensional sparse vector.
- Validation Strategy: Stratified 80/20 split plus 5-fold StratifiedKfold cross-validation, ensuring proportional class representation in each fold given the severe class imbalance.
- Model Selection: The model with the highest macro F1 on the validation set is selected as the final model.

V. EXPLORATORY DATA ANALYSIS

A. Data Quality

Completeness

The analysis of missing values reveals a critical pattern in diagnosis columns: while the primary and first secondary diagnoses are almost completely populated, the information density drops drastically in later columns. Columns beyond Diagnosis 26 exceed 98.9% missing values, reaching 99.81% in column 35. This pattern indicates that clinically useful information is effectively concentrated in the first 10-15 diagnosis and procedure fields.

Diagnosis Column	% Missing
Diag 26 Secondary	98.94%
Diag 27 Secondary	99.11%
Diag 30 Secondary	99.48%
Diag 33 Secondary	99.70%
Diag 35 Secondary	99.81%

Table I. Columns with Highest Percentage of Missing Data

Correctness

ICD-10 and ICD-9-CM codes show high consistency with the master reference catalogs. The sex variable and age field require preprocessing to eliminate possible format inconsistencies. All DRG codes in the target column were verified against the official IR-GRD V3.1 catalog. Regex validation confirms that more than 95% of codes have the correct format according to ICD-10 standards (pattern `[A-Z][0-9]{2}`) and ICD-9-CM (pattern `[0-9]{2}.[0-9]`).

Outliers

The age variable has values up to 121 years which, while biologically possible, are extreme values that could bias non-robust models. The applied solution was to clip age to the observed 1st-99th percentile range and impute missing values with

the median (39 years). The distribution also shows a high concentration of pediatric patients (0-5 years), consistent with a hospital serving obstetric and neonatal populations. The dataset also recorded 0 exact duplicates.

B. Descriptive Statistics

From the 14,561 patient records, the following population characteristics were identified:

Variable	Value	Detail
Total records	14,561	Hospital discharges
DRG classes	526	Multiclass target variable
Mean age	39.4 years	SD = 24.7 years
Median age	36 years	IQR: 23-60 years
Female patients	9,617 (66%)	Majority class
Male patients	4,944 (34%)	

Table II. Descriptive Statistics of the Dataset

The marked female predominance directly reflects the composition of the dataset: the most frequent DRG categories are obstetric and neonatal (cesarean sections, vaginal deliveries, antepartum disorders, neonates). The DRG distribution is highly skewed: the top class (146101, Cesarean section) has 813 records, while approximately 200 classes have 5 or fewer records — the main classification challenge.

C. Relevant Feature Visualizations

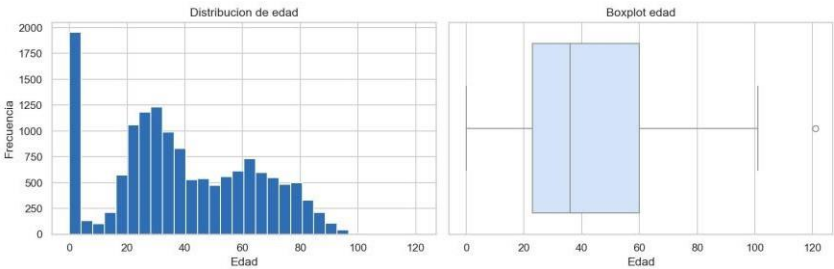


Fig. 1. Age distribution histogram (left) and boxplot (right). IQR: 23–60 years; outliers above 100 years are observed.

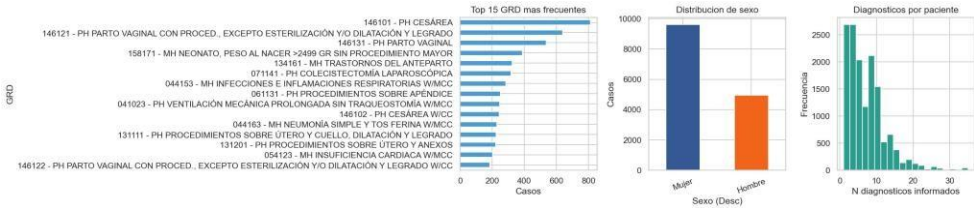


Fig. 2. Top-15 most frequent DRGs (left), sex distribution (center), and number of diagnoses per patient (right).

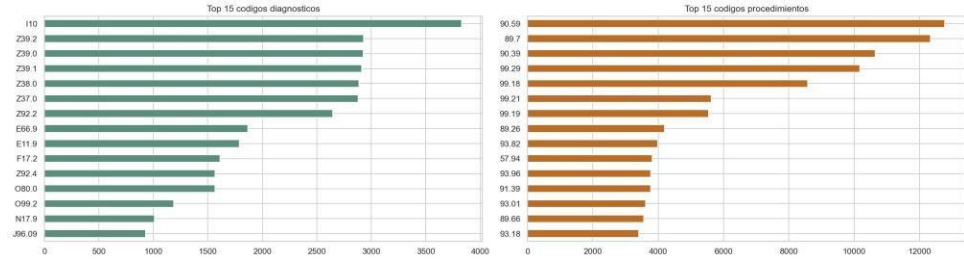


Fig. 3. Top-15 ICD-10 diagnosis codes (left) and Top-15 ICD-9-CM procedure codes (right). Obstetric and postpartum codes (Z39.x, Z38.x) dominate, consistent with the DRG distribution.

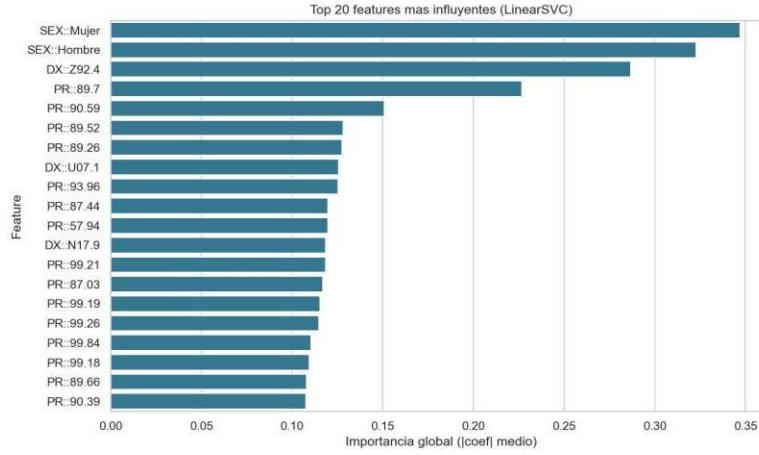


Fig. 4. Top-20 most influential features in the LinearSVC model, measured as the mean absolute coefficient value across all One-vs-Rest (OvR) classifiers.

VI. EXPERIMENTS

A. Feature Selection Justification

The selected input features (X) are: ICD-10 diagnosis codes, ICD-9-CM procedure codes, patient age and biological sex. This selection is grounded in both clinical and technical reasoning.

From a clinical standpoint, these four variable types constitute exactly the information used by the official IR-GRD grouper software to assign a DRG. No other structured variables with equivalent predictive power are available in the administrative dataset. The target variable (y) is the DRG code, representing clinical severity and resource consumption.

From a technical standpoint, since each patient may have multiple diagnosis and procedure codes without a defined clinical ordering, treating them as an unordered set avoids spurious positional dependencies. The MultiLabelBinarizer transforms this unordered set into a binary presence/absence vector across all observed codes, capturing co-occurrence patterns without ordering bias.

Feature importance analysis (Fig. 4) confirms that sex (female/male) has the highest discriminative weight, followed by specific diagnosis codes (Z92.4 - long-term prophylactic medication; U07.1 - COVID-19; N17.9 - acute renal failure) and procedure codes (89.7 - respiratory monitoring; 90.59 - laboratory tests; 57.94 - urinary catheterization). The dominance of sex as a feature is explained by the predominance of obstetric DRGs in the dataset, which are exclusive to female patients.

B. Training Results and Model Selection

Three models were evaluated under identical conditions. Table III summarizes their performance on the validation set and computational cost:

Model	Macro F1	Weighted F1	Accuracy	Time (s)
LinearSVC (selected)	0.328	0.626	64.2%	7.9
Logistic Regression	0.280	0.544	54.4%	225.9
Dummy (most frequent)	0.000	0.006	5.6%	0.007

Table III. Model Comparison on the Validation Set

LinearSVC was selected as the final model for two reasons: (1) it achieves the best macro F1 (+17% relative improvement over Logistic Regression), indicating superior handling of minority classes; and (2) it is 28.5x faster to train than Logistic Regression (7.9 vs. 225.9 seconds) — an important consideration for periodic model retraining in a hospital information system. Fig. 5 compares training and cross-validation times across models.

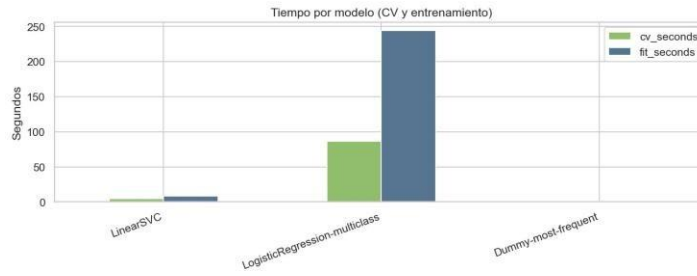


Fig. 5. Comparison of training and cross-validation times across models. LinearSVC achieves better performance at a fraction of the computational cost.

C. Model Architecture

The final model follows a parallel preprocessing pipeline that handles the heterogeneous nature of clinical data (Fig. 6). The pipeline has four stages:



Fig. 6. Model architecture: parallel preprocessing branches for clinical codes and demographic variables, assembled into a sparse feature vector and classified using LinearSVC.

(1) Clinical Branches — MultiLabelBinarizer transformations applied independently to diagnosis and procedure code sets, generating high-dimensional sparse binary matrices; (2) Demographic Branch — OneHotEncoder for sex and median-imputed numeric age; (3) Feature Assembly — sparse horizontal concatenation of all branches into a unified vector; and (4) Classifier — LinearSVC with One-vs-Rest strategy for the 526-class problem.

D. Performance Analysis

On the full test set (526 classes), the model achieves the following final metrics:

Metric	Value
Global Accuracy	62.9%
Macro F1	0.290
Weighted F1	0.611
Top-3 Accuracy	76.9%

Table IV. Final Model Metrics (526 DRG Classes)

The large gap between macro F1 (0.29) and weighted F1 (0.61) is a direct consequence of class imbalance: hundreds of DRG categories with very few training examples are systematically misclassified. For the top 20 most frequent classes, performance improves substantially to a weighted F1 of 0.85 (Table V).

DRG (Top Selected)	Precision	Recall	F1
146101 - PH CESAREAN SECTION	0.93	0.96	0.94
146121 - PH VAGINAL DELIVERY W/ PROC.	0.93	0.98	0.95
146131 - PH VAGINAL DELIVERY	0.89	0.98	0.93
041023 - PH PROL. MECH. VENT. W/MCC	0.94	0.92	0.93
131111 - PH UTERUS & CERVIX PROC. D&C	0.88	0.98	0.93

DRG (Top Selected)	Precision	Recall	F1
061131 - PH APPENDIX PROCEDURES	0.88	0.98	0.92
071141 - PH LAPAROSCOPIC CHOLECYSTECTOMY	0.79	0.92	0.85
054123 - MH HEART FAILURE W/MCC	0.50	0.44	0.47
Micro Average (Top-20 classes)	0.85	0.86	0.85

Table V. Classification Report — Top-20 Selected Classes

The confusion matrix (Fig. 7) for the 12 most frequent classes shows a strong diagonal dominance, indicating the model rarely confuses the main obstetric categories with each other. The only notable off-diagonal entry is for DRG 146102 (Cesarean with CC), which is partially confused with DRG 146101 (Standard Cesarean) — a clinically intuitive error given the similarity of their coding profiles.

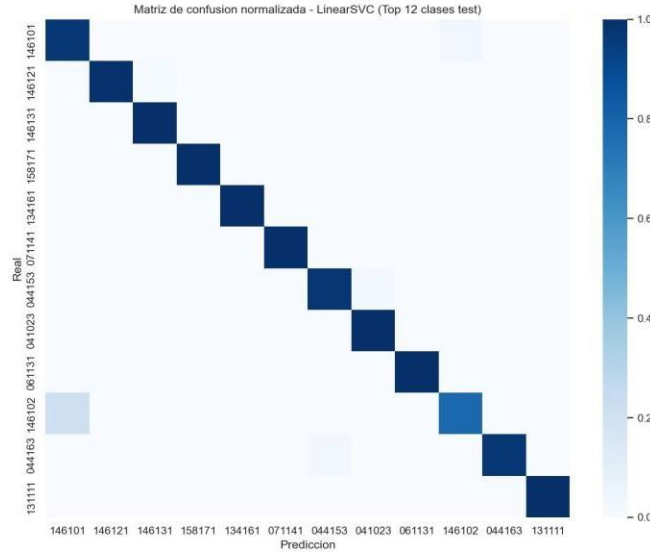


Fig. 7. Normalized confusion matrix for the 12 most frequent DRG classes. The strong diagonal pattern confirms high accuracy in the most frequent categories.

E. Comparison with Literature

Table VI contextualizes our results within the international DRG prediction literature:

Study	Model	Classes	Accuracy	Note
Goldstein et al. [3]	Decision Tree	~100	~70%	Subset
Shi et al. [4]	CNN + Embed.	~200	F1-W: 0.78	Free text
Moons et al. [6]	LinearSVC TF-IDF	312	68%	Codes only
This work	LinearSVC (OvR)	526	62.9% / Top-3: 76.9%	All classes

Table VI. Comparison with International DRG Prediction Literature

Our global accuracy of 62.9% and Top-3 of 76.9% are competitive given that our task covers 526 classes — almost double the most comparable study (Moons et al. with 312 classes). For the high-frequency subset (top-20 classes, weighted F1 = 0.85), results surpass most reported benchmarks even for studies using deep learning on free text.

VII. CONCLUSIONS

This study demonstrates that a LinearSVC classifier with a sparse multi-hot feature representation of ICD codes is a viable and computationally efficient approach for DRG prediction from structured clinical data. On the Hospital El Pino dataset with 14,561 records and 526 target categories, the model achieves 62.9% global accuracy and Top-3 accuracy of 76.9%, meaning that in more than 3 out of 4 cases the correct DRG appears among the model's top 3 suggestions.

The model excels in high-frequency categories, particularly obstetric DRGs, achieving weighted F1 above 0.85 for the top 20 classes. This confirms the practical utility of the system as a coding assistance tool for the most common types of hospital discharge.

The dominant features are biological sex and a small set of highly informative diagnosis and procedure codes, which aligns with clinical expectations. The fact that sex has the highest discriminative weight reflects the dataset structure where obstetric cases dominate.

The main limitation is the macro F1 of 0.29, driven by the hundreds of DRG categories with very few training examples. Addressing class imbalance will be essential for a production-ready system capable of reliably classifying rare conditions.

VIII. LIMITATIONS AND FUTURE WORK

The limitations of the current work suggest the following improvement directions:

- **Class Imbalance:** The large number of rare DRG categories limits macro F1. Future work should explore oversampling, class-weighted losses, or hierarchical classification.
- **Rare Class Filtering:** Grouping or excluding DRG classes with very few examples (e.g., < 10 records) into an 'other' category could substantially improve both macro and weighted F1 without significantly reducing clinical utility.
- **Code Embeddings:** Replacing the binary MultiLabelBinarizer with pre-trained ICD code vector representations (e.g., Med2Vec or clinical BERT models) could capture semantic relationships between codes and improve generalization for rare categories.
- **Ensemble Methods:** A two-stage system combining LinearSVC (as a fast first-stage filter) with a gradient-boosted tree (e.g., LightGBM) for difficult cases could improve macro F1 while maintaining computational efficiency.
- **Temporal Validation:** The current evaluation uses a random split. In production, a temporal split (train on older records, test on recent ones) would provide a more realistic estimate of performance as coding practices evolve.
- **Multi-hospital Generalization:** The model was trained and tested exclusively on Hospital El Pino data. Future work should evaluate transferability to other FONASA-affiliated hospitals with different case-mix distributions.

IX. PHASE 2 — CHATBOT: SPORTS DOMAIN

As part of Phase 2 of the project, our group has selected Sports and Athletic Performance as the knowledge domain for the development of an intelligent chatbot using an open local LLM (Llama 3 or DeepSeek).

A. Domain Description and Justification

Sports and Athletic Performance encompasses the study of training, physical conditioning, sports rules, athlete performance, and competition across a wide range of disciplines. This domain is relevant and accessible to a broad audience because: (1) sports are among the most widely followed topics worldwide, with millions of fans and practitioners seeking reliable information; (2) athletes, coaches, and enthusiasts regularly need accurate information about training methods, sports regulations, and performance strategies; and (3) sports knowledge spans well-defined, verifiable domains making it ideal for structured Q&A.

A chatbot capable of answering sports-related questions would serve athletes, coaches, students, and fans by providing quick access to accurate information about rules, training, and performance. The domain is well-suited for LLM-based Q&A because: it has a rich and well-documented knowledge base (official rulebooks, training science literature, sports statistics), the questions tend to be clearly defined, and answers can be grounded in reliable sources without requiring real-time data.

REFERENCES

- [1] R. B. Fetter, Y. Shin, J. L. Freeman, R. F. Averill and J. D. Thompson, "Case mix definition by diagnosis-related groups," *Medical Care*, vol. 18, no. 2, pp. 1-53, 1980.
- [2] 3M Health Information Systems, "All Patient Refined DRGs (APR-DRGs) Methodology Overview," 3M, Technical Report, 2003.
- [3] I. Goldstein, H. Arzoni and S. Ben-Assuli, "Predicting hospital readmission and DRG classification using machine learning on EHR data," *Journal of Medical Systems*, vol. 44, no. 3, pp. 1-9, 2020.
- [4] T. Shi, S. Chakraborty and P. Srihari, "DRG prediction from ICD-10 codes using convolutional neural networks," in *Proc. AMIA Annual Symposium*, pp. 1732-1741, 2021.
- [5] T. Baumel, J. Nassour-Kassis, R. Cohen, M. Elhadad and N. Elhadad, "Multi-label classification of patient notes: Case study on ICD code assignment," in *Proc. AAAI Workshop on Health Intelligence*, 2018.

- [6] P. Moons, W. Budts and J. De Geest, "Critique on the conceptualisation of quality of life and automated DRG grouping from ICD codes using supervised classification," *Health Informatics Journal*, vol. 26, no. 4, pp. 2923-2937, 2020
- [7] M. Soto, C. Villarroel and M. Araneda, "Calidad del codificado GRD en hospitales chilenos: analisis de discordancias," *Revista Medica de Chile*, vol. 148, no. 7, pp. 955-963, 2020.
- [8] T. Joachims, "Text categorization with support vector machines: Learning with many relevant features," in *Proc. 10th European Conference on Machine Learning (ECML)*, pp. 137-142, 1998.

Link GitHub: <https://github.com/iclxme/Aprendizaje-Maquina.git>